MAPPA: Symmetry-Aware Atom Mapping through Continuous Fragment Growth

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Atom-to-atom mapping traces atom identities from reactants to products, supporting reaction analysis and pathway construction. Yet the same endpoints can admit multiple mappings, and molecular symmetry makes explicit exploration of these alternatives expensive. We introduce MAPPA, a constraint-based framework that efficiently discovers and organizes alternative atom mappings. MAPPA combines greedy growth of shared structural fragments with a systematic bond-cut sweep to expose correspondences that individual growth trajectories can miss. Symmetry-related assignments are represented compactly, and a separate decoder reuses fragment calculations to organize alternatives under an approximate bond-event model. This separation supports parallel exploration and preserves alternatives for subsequent chemistry-aware selection. MAPPA achieves the highest reference coverage among the evaluated methods on Golden, with decoded reference recovery reaching 99.46% on Golden and 99.63% on EnzymeMap. With one seed, mapping and decoding together average 3.19 CPU seconds per Golden reaction and 13.16 CPU seconds per EnzymeMap reaction.

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1 Introduction

Atom-to-atom mapping identifies corresponding atoms across a chemical reaction. It describes the net evolution of atom identities from reactants to products: the endpoint correspondence left by a chemical pathway, rather than the sequence of intermediates or elementary steps along that pathway. These correspondences support reaction-center identification, template extraction and the preparation of structures for reaction-path calculations. Yet endpoint structures do not always determine a unique assignment. Different correspondences may imply different atom origins and bond changes, while molecular symmetry can produce many equivalent assignments.

Two complementary objectives organize AAM methods by their intended output. *Alternative-generating solvers* retain several admissible correspondences, allowing a later decision between them. SLAPMapper, which combines label refinement with sequential linear assignment, belongs to this category (9). *Prediction-oriented solvers* select a most plausible correspondence: RXNMapper derives an assignment from reaction-language-model attention (14), while LocalMapper learns from chemist-labeled reactions (1). These are output objectives, not a strict separation between optimization and learning. Reference inclusion anywhere in a candidate set measures coverage; the correctness of one selected answer measures top-1 accuracy.

An alternative-generating solver can support a single-answer workflow through a separate selector. A simple surrogate favors fewer bond events; chemical rules, learned preferences or reaction context could then distinguish the remaining choices. Such a selector cannot recover a correspondence absent from its input set, and its accuracy must be evaluated separately. MAPPA belongs to the alternative-generating category, as does SLAPMapper. Its purpose is to expose and organize structural choices without repeatedly listing permutations of the same choice; we do not introduce or evaluate a chemistry-trained selector here.

Weighted molecular graphs introduce another distinction. Two assignments can both satisfy a permissive bond-weight matching tolerance while producing different bond events at a finer threshold. Retaining one representative alone can therefore hide information needed for chemical interpretation. The search for correspondences and their interpretation as bond changes should remain separate.

MAPPA separates three decisions: whether a fragment can continue growing, which compressed placements its completed constraints permit, and what bond changes those placements imply. Growth retains one compatible assignment. At saturation, a separate operation constructs the distinct compressed fragment placements, which become conditional branches for mapping the remaining atoms. A cut sweep varies the growth conditions to expose alternatives excluded by greedy fragment commitment. The completed search is reduced to a catalogue of mapping families, and a separate decoder shares local bond-event calculations across its unique fragments. We describe these stages and evaluate reference recovery and computational cost on Golden (10) and EnzymeMap (7).

2 Methods

2.1 Atom mapping as constraint discovery

We treat atom mapping as a constrained search for correspondences shared by the two endpoints. Element identity and one-to-one occupancy restrict assignments. A compatible fragment adds a stronger structural constraint: several atoms and their accepted connections must coexist in the reactant and product. Locking that fragment conditions the search for the remaining atoms. Symmetry retains the alternative assignments that are still consistent with that shared structure.

Greedy growth discovers how far a constraint can extend under the current assignments and growth trajectory. The resulting fragment is *locally saturated*, not necessarily the largest common fragment attainable under every trajectory. Large fragments can restrict many atom choices at once, while avoiding unnecessary early commitments to short repeated motifs. Different seeds and frontier choices can discover different saturated constraints, which is why additional seeds can improve coverage.

This preference also creates a bias. In the schematic transformation of Figure 2a, a large B–A region can dominate seed selection and repeatedly lock as the same preserved unit, followed by R–B. If that region

contains 90% of the eligible seed atoms, a uniformly selected initial seed lies there with probability 0.9. Repeating seeds can then revisit a similar constraint decomposition. A single-edge sweep systematically masks each eligible preservation edge in turn, including an edge inside the dominant region, to expose growth trajectories that ordinary seeding may rarely reach. The mask does not assert that the bond breaks in the reaction. This exploration principle can also be applied to another mapper; our expanded SLAP experiment reaches approximately 97% coverage (Section 4.4).

2.2 Scientific stages and module boundaries

MAPPA separates correspondence discovery from bond-event interpretation. Matching returns conditional fragment placements. Reduction collects them into a catalogue shared across seeds and cuts. Decoding prepares reusable fragment responses and returns branches that can be queried within an event-count window (Figure 1). These stages have separate inputs and outputs: decoding a saved catalogue does not repeat matching.

A *fragment* pairs a source atom set with a target atom set and retains one original discovery record. A *witness* is one explicit correspondence permitted by that record. A *branch* is a composition of fragment choices; a *mapping family* retains their selected local relations and ordered placement context. Symmetry describes correlated rearrangements within these records, rather than a list of complete atom permutations.

The implementation follows these scientific boundaries. A shared core supplies binary symmetry, constrained assignment and event-response operations. Matching, reduction and decoding consume that core through distinct stage interfaces. Execution and persistence manage work and intermediate results; they do not decide which mappings are allowed. Reference verification is an external evaluation of the saved outputs (Section 3.2).

2.3 Endpoint graphs and discovery constraints

Let V_R, V_P be the endpoint atom sets, Z_R, Z_P their elements and W_R, W_P their bond-weight matrices. A partial mapping $m : D \subseteq V_R \to V_P$ is injective and element-preserving. Explicit hydrogens participate. Coordinates may accompany the inputs, but do not determine growth.

The active source graph contains bonds at or above the presence threshold b. A growing fragment stores only its accepted preservation edges A. For each $(u, v) \in A$, discovery requires

$$Z_R(u) = Z_P(m(u)), \quad W_P(m(u), m(v)) \geq b, \quad |W_R(u, v) - W_P(m(u), m(v))| \leq \tau_{\text{iso}}. \tag{1}$$

Previously completed assignments remain locked during subsequent growth. Other bonds between included atoms are not implicitly added as constraints, and source nonedges impose no negative constraint. The complete original matrices are retained for event interpretation. Thus the accepted growth graph can differ from the fragment's final internal connectivity.

2.4 One-witness fragment growth

Starting from a seed, MAPPA selects a boundary atom uniformly and then one of its incident frontier edges uniformly. Each proposal asks a single question: does the enlarged fragment have at least one compatible injective assignment? Bond weight does not rank proposals.

The current witness is extended when possible. Otherwise, a previously discovered fragment can certify existence under the current occupancy and edge constraints (Section 2.10). If neither supplies a witness, the existence solver propagates element, bond-support and distinct-target constraints before searching the remaining choices. Assignments within the growing fragment may change; completed-fragment locks remain fixed. The query stops at the first witness. Only exhaustion establishes absence, so a resource limit cannot become a rejected chemical match.

A successful proposal adds its edge and replaces the witness. An impossible proposal is deferred while other frontier edges are considered. When a completed fragment is joined, its locked assignments and accepted edges are retained. Saturation occurs when no admissible extension remains. Intermediate growth therefore does not enumerate all matches of each shorter fragment.

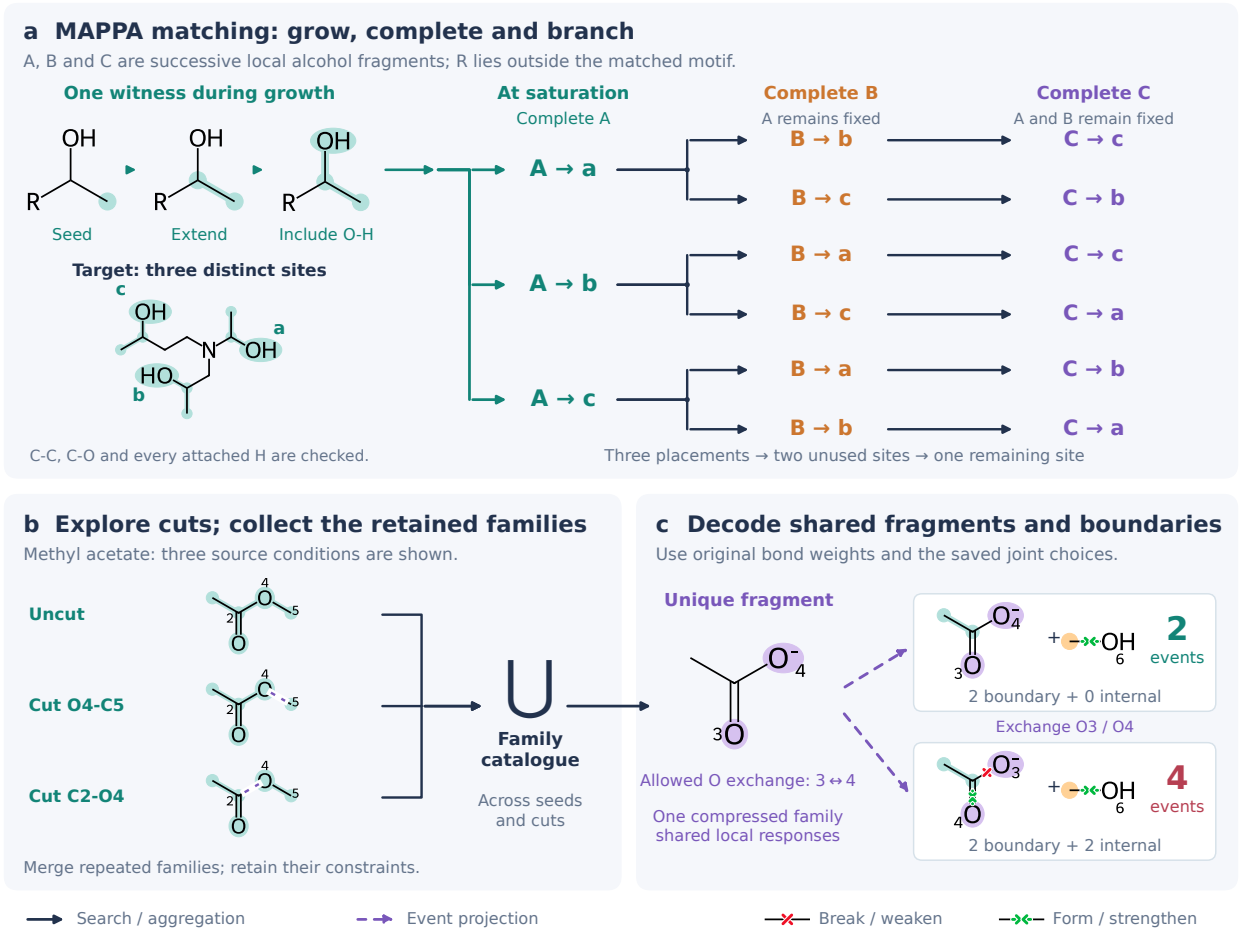


Figure 1 From fragment matching to bond-event interpretation. (a) Growth keeps one witness; saturation produces the compressed placements used for conditional branching. Three illustrative source alcohol motifs can occupy three target sites with distinct molecular environments. Placing A conditions the sites available to B and C. Attached hydrogens participate in matching but are implicit in the drawings. (b) Uncut and single-edge-masked ester searches contribute families to one catalogue across seeds and cuts. A mask changes the growth conditions, not the bond events. (c) Decoding shares internal responses for unique fragments, with a fixed ordinary-boundary contribution and an endpoint-dependent sweep correction. A permitted binary-symmetry exchange of the acetate oxygen images adds two internal changes to two boundary changes at event threshold 0.5. Black lines show endpoint bonds, red × marks show cleavage or weakening, and green inward arrows show formation or strengthening. Dashed arrows project choices within the compressed family. Panel a is a constructed local-placement example; panels b and c use methyl acetate saponification. These illustrations are not a complete search trace, and the formal-bond-order signatures are not distinct mechanisms or resonance-normalized transformations.

2.5 Compressed completion and the two roles of symmetry

At saturation, a separate operation discovers the allowed target atom sets for the completed source fragment. It uses the full accepted constraints and exclusions for placements already represented, rather than emitting every internal bijection and merging duplicates. A six-membered carbon ring mapped onto one equivalent ring gives one atom-set placement; two rings with distinct surrounding environments can give two placements. Their internal rotations do not create additional atom-set candidates.

Symmetry has two roles. *Fragment-local symmetry* is computed on each fragment's own binary internal graph, preserving elements and fixing its anchors. Bonds above the presence criterion have one structural label; original weights remain available for event counting. Nauty provides compact generators (11), each describing a permitted joint rearrangement. For example, exchanging two identical arms moves their attached

atoms together; it does not allow independent atom choices that break connectivity. Both source and target fragment actions are retained. If an aligned pair has equal binary connectivity, one side suffices to represent their relative rearrangements.

Global placement symmetry acts on an entire endpoint and can connect different target atom sets. To share a subsequent search continuation, MAPPA uses only a certified subgroup that preserves the target weights, active connectivity, source-to-target compatibility and earlier locks. The saved action moves the whole continuation jointly. This stronger requirement protects later growth; it is distinct from the permissive local binary symmetry used in decoding. If global equivalence is broken by external attachments, the local fragment can still have its own symmetry.

A fragment stores its atom-set pair, original witness, discovery constraints, anchors and local symmetry relations. Different discovery correspondences on the same sets need not belong to one binary-symmetry orbit. This distinction matters when decoding selects a starting correspondence (Section 2.8); neither a single witness nor individual atom domains alone describe every correlated choice.

2.6 Hierarchical branching, pruning and cut sweep

Each completed placement creates a continuation for the remaining atoms. Earlier choices condition later ones (Figure 1a), and targets cannot be occupied twice. After each finished search step, the hierarchy is recounted from its leaves upward. At an over-limit node, the largest child subtree is removed, the node is recounted, and pruning continues until its retained count is within the branch cap. An ancestor uses the already-reduced descendant count. The cap counts representative continuations, not the symmetry-related bijections within a fragment. Truncation remains explicit.

Greedy growth can commit to constraints that exclude another useful correspondence. MAPPA varies these commitments through independent seed orderings and a single-edge cut sweep. For $G_R = (V_R, E_R)$, the sweep searches the uncut graph and $G_R^{(e)} = (V_R, E_R \setminus \{e\})$ for each eligible edge, including bonds to hydrogen. Each seed setting supplies that many source-atom orderings per cut condition.

A masked edge changes growth, not the original endpoint weights. Its endpoints may reconnect through another route, and the final mapping may preserve that bond. Conversely, one mask can redirect several later choices and expose multiple bond changes. Both reaction directions are searched and retained in their own atom-index frames. Seeds, cuts and bidirectional search broaden exploration, but do not make it exhaustive.

Reactions and directions can be processed independently. Within a direction, independent seed/cut searches can run concurrently and feed the same later reduction. Growth decisions along one conditional branch remain sequential: later fragments depend on earlier occupancy. Parallel execution changes where these operations run, not their constraints or random streams.

2.7 Atom-set reduction and the mapping catalogue

Reduction identifies each completed fragment by the pair (R_i, P_i) of source and target atom sets. It retains one original fragment record per pair and gives it a shared identifier. A final branch is grouped by its unordered composition

$$\mathcal{B} = \{(R_1, P_1), \ldots, (R_q, P_q)\}. \tag{2}$$

Repeated witnesses or growth orderings do not create new fragment identities. Branch compositions also retain their ordered global placement context: overlapping actions must remain correlated, whereas independent decisions may be reordered.

This is a structural aggregation. It reuses the inventories accumulated during matching, translates compositions to shared fragment identifiers and deduplicates those compositions. It performs no symmetry solving, starting-map optimization or event counting. The retained original record keeps its witness, domains, accepted edges and anchors intact. Duplicate discovery records are discarded rather than combined into a broader relation. Consequently, reduction selects a representative discovery record; it does not preserve the union of every discarded starting relation. Alternative partitions and cut histories remain compact provenance

Algorithm 1 Matching, reduction and separate decoding

Require: Endpoint graphs, discovery policies, seed/cut plan, branch cap and event window
for all requested search directions **do**
  for all cut conditions and seed orderings **do**
    Initialize the conditional search hierarchy
    while a retained continuation has an eligible fragment seed **do**
      Grow using one-witness existence queries
      At saturation, construct distinct compressed placements
      Continue each placement with earlier assignments locked
      Prune and recount the hierarchy from leaves to root
    end while
    Retain terminal compositions and compact provenance
  end for
  Share fragments by atom-set pair; reduce repeated branch compositions
end for
Complete the matching/reduction stage before batch decoding
for all reaction/direction catalogues **do**
  Inventory unique fragments and prepare their shared symmetry-event responses
  Compose branch contributions and the sweep-endpoint correction
  Retain queryable branches within the requested event window
end for

for interpreting the retained families. One complete reaction/direction catalogue, rather than an isolated cut, is the input to decoding.

2.8 Shared fragment symmetry-to-event responses

The decoder aims to reuse independent fragment calculations instead of enumerating whole-reaction bijections. Its work has three parts: identify unique fragments, discover their internal symmetry-dependent events, and assemble those responses with outer boundary contributions. Fragment independence is a design objective and an explicit boundary approximation, not a consequence of atom-set disjointness alone. Events use the original endpoint weights:

$$\Delta_{uv}(m) = W_P(m(u), m(v)) - W_R(u, v). \tag{3}$$

A change is counted when $|\Delta_{uv}|$ is strictly greater than 0.5, or 0.3 for a pair involving a metal. Signs distinguish formation or strengthening from cleavage or weakening. These event thresholds are separate from discovery tolerance; binary rearrangements can preserve connectivity while changing weighted bond events.

For each unique fragment, decoding retains one starting correspondence and its permitted local binary symmetries. It first improves the saved witness under those symmetries. Additional starting-correspondence search is limited to fragments whose accepted growth connectivity differs from their final connectivity on at least one side, and whose current witness remains above an established lower bound. This work has a fixed per-fragment budget. An attained bound certifies the minimum within its stated scope; otherwise the result is the best witness found with incomplete optimality. Selecting one starting orbit is an explicit restriction on decoded coverage.

The A-substituted C_3 ring in Figure 2b illustrates why completed connectivity matters. Growth along C_1–C_2–C_3 places the A-bearing C_1 at a target-chain end, preventing extension to A, which is attached at the target center. The completed carbon triangle nevertheless permits an internal rearrangement moving C_1 to that center. This preserves the A attachment while leaving the internal ring-opening contribution unchanged. For this small example, local symmetry already supplies the displayed alternatives. More general unequal-connectivity fragments can have useful starting correspondences outside one symmetry orbit; the bounded search attempts to improve that starting position without enumerating all such orbits.

The shared response contains correlated symmetry operations, a fixed contribution and sparse variable bond-event terms. Independent components are separated only when neither an event term nor a requested endpoint couples them. Event-invariant operations can share computation. Occupancy remains joint: two arms cannot

choose the same target position. The decoder retains this compact relation instead of materializing every permutation or every event total. A later query asks for an assignment at a selected count or with selected atom pairs and returns a replayable witness, absence or an unresolved outcome.

2.9 Branch composition, sweep correction and event windows

Each branch references the prepared local responses. Its reported count uses the independent ordinary-boundary model

$$E_{\mathcal{B}}(m) = B_{\mathcal{B}} + \sum_i e_i(m) + s_{\mathcal{B}}(m), \qquad 0 \le s_{\mathcal{B}}(m) \le 2. \tag{4}$$

Here e_i counts internal changes owned by a disjoint final fragment piece, $B_{\mathcal{B}}$ is the fixed ordinary-boundary contribution, and $s_{\mathcal{B}}$ is the correction associated with the single swept bond. Mapped/unmapped boundaries contribute without inventing correspondences for unmatched atoms; bonds wholly within unmatched material are omitted. Alternative partitions or cut histories are alternative compositions, not additive contributions.

The sweep correction depends on its two endpoints and on local choices that move them. Assembly reuses the prepared fragment responses and preserves the joint endpoint/internal-event relationship. Shared placement actions remain one choice across the continuation. Ordinary boundaries, however, are not recomputed for every rearrangement. Equation (4) is therefore an approximation to the physical whole-reaction event count, not a claim that all inter-fragment changes are independent.

Greedy growth is important to making this approximation useful: a connection that can still be absorbed should be incorporated before a fragment locks. A restrictive discovery tolerance can create an avoidable boundary. In the heavy-atom illustration C–C≡N → C–C–N (Figure 2c), tolerance one rejects matching the triple bond to the single bond. If the C–C fragment is left separate from N, exchanging its two carbons changes the boundary events from one weakening to a cleavage plus a formation. Tolerance two permits this connection to be absorbed, so its event belongs to the internal fragment response. Permissive greedy growth reduces such artificial boundaries, but cannot guarantee independence: the ring example retains a dependence caused by differing connectivity. The model keeps ordinary boundaries fixed and treats the swept connection jointly; it does not silently solve every ordinary-boundary correlation.

Window selection combines local lower bounds with attained branch witnesses. If attained local minima compose, the smallest pre-sweep composition L has an attained upper bound at most $L+2$. Only compositions that can improve the current best count need the corresponding refinement. If local optimization is incomplete or its witnesses cannot compose, the decoder instead uses the honest lower bound and an actual attained upper bound U; $L + 2$ is then not assumed to be attainable. Branches whose lower bounds exceed $U + n$ are excluded, and the survivors retain their event-query responses. The studies below use $n = 5$. A proved model minimum, a best-found model count and a physical event minimum remain distinct.

2.10 Parallel execution and shared computation

Matching can reuse a previously discovered fragment as an existence witness when its restriction satisfies the proposed edges, locks and current target availability. Every seed and cut still makes fresh growth decisions. This implementation detail does not change the scientific search objective; the measured seed scaling is reported separately.

The same stage interfaces support direct local calls and persistent worker pools. A pool is created once and schedules reaction, seed or cut work without putting scheduler decisions inside the algorithms. For a large campaign, all matching and reduction finish before decoding begins. Intermediate records can remain in memory or pass through aggregated storage with shared immutable fragments; file counts do not grow with individual seeds or cuts. Compatible reaction-cache snapshots share useful fragment proofs across workers. Retention limits affect reuse, not the declared mapping relation.

Reduction is local to a reaction and direction. Once its complete catalogue is available, that catalogue can be decoded independently of other reactions. Within it, unique-fragment response preparation is separable when its inputs are fixed; branch assembly must retain shared placement and sweep-endpoint dependencies. The reported execution pools distribute catalogue decoding by reaction/direction rather than launching a

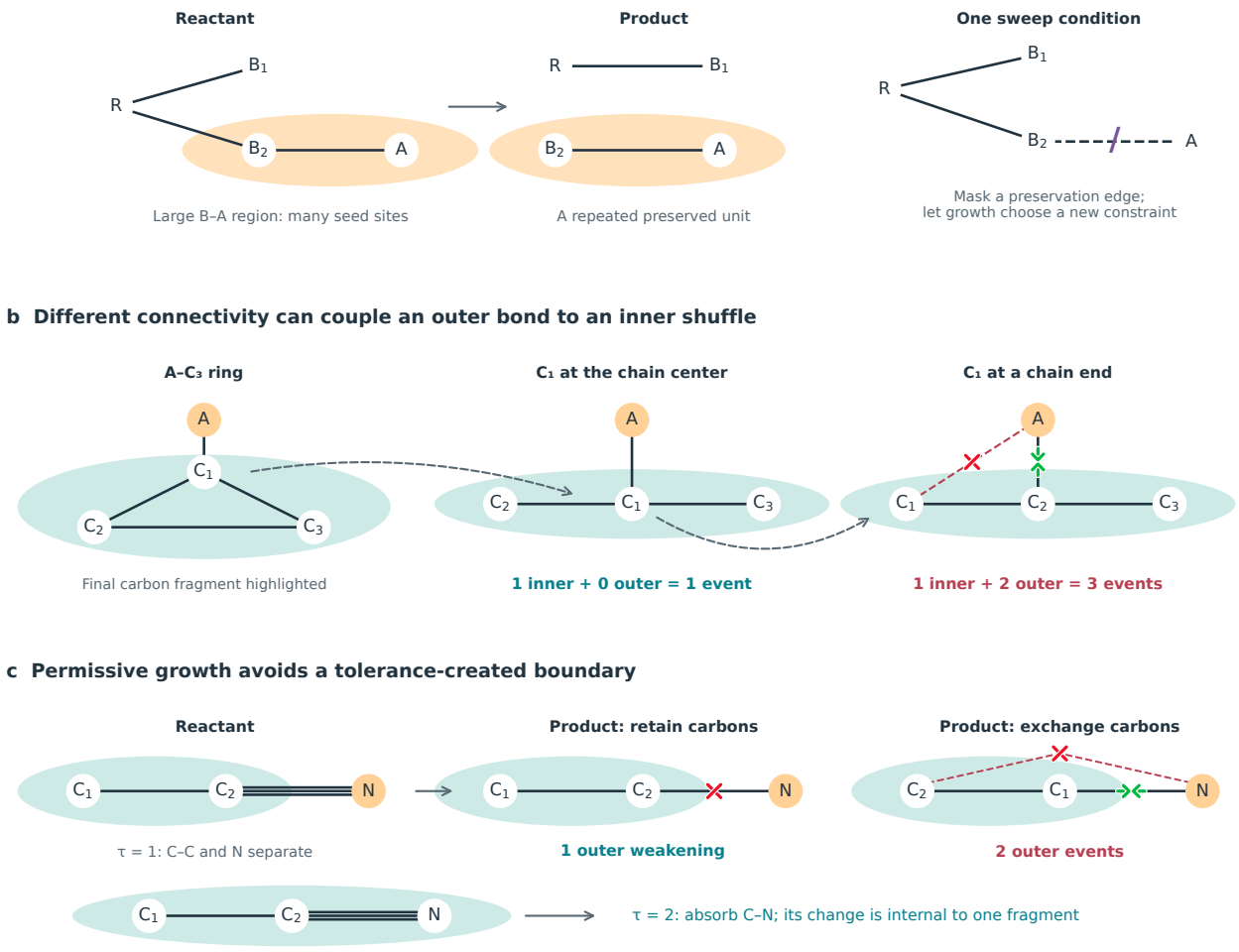


Figure 2 Structural constraints guide exploration and decoding. (a) A large preserved B–A region can dominate greedy growth; masking one of its preservation edges perturbs that commitment. R, A and B denote schematic groups, not measured reaction frequencies. (b) An A-substituted carbon triangle maps to an A-substituted carbon chain. Both displayed carbon assignments have one internal cleavage. Placing the A-bearing carbon at the chain center preserves the outer attachment; placing it at an end adds a cleavage and a formation. The physical totals are one and three, respectively. Dashed correspondence lines follow the same indexed atoms. (c) On the displayed heavy-atom graphs, tolerance one can leave C–C and N separate when a triple bond becomes single. A carbon exchange then changes outer events from one to two. Tolerance two allows the connection inside one fragment. Panels b and c are analytically checked graph examples with unit bond orders except the indicated triple bond, event threshold 0.5, and no hydrogen constraints. They explain the independence approximation and do not assert a measured failure frequency. The visual bond-event symbols follow Figure 1.

task for every fragment. Thus the useful parallel boundaries follow scientific dependencies, with shared local work retained inside each catalogue.

Bulk assignment, composition and event operations use native kernels, while the public records and stage interfaces remain in Python. The shared symmetry/event core serves both local fragment preparation and branch decoding. Persistence stores the compact responses and chosen starting correspondences, so loading a decoded result does not repeat optimization. Detailed query, storage and cache rules appear in the supplement.

3 Evaluation

3.1 Datasets and search settings

We evaluate all 1,851 Golden reactions (10) and the prepared collection of 47,974 EnzymeMap reactions (7). Reference labels are removed before search. Both studies use the prepared formal-bond-order matrices and explicit hydrogens; these results do not constitute a benchmark of continuously valued electronic-structure WBO inputs. EnzymeMap references are curated database mappings, so agreement is not independent validation of reaction mechanisms.

The common settings are discovery tolerance $\tau_{\text{iso}} = 2$, presence/cut threshold $b = 0.2$, hierarchical branch cap 100, random seed 42, both directions and the full single-edge sweep. Golden uses one and ten seed orderings per cut; EnzymeMap uses one. Decoding uses the original weights, thresholds 0.5/0.3, a five-second allowance per unique fragment and the best attained model count plus five. No reference mapping guides growth, pruning or starting-correspondence selection.

A separate paired Golden study evaluates current one-seed MAPPA and pinned SLAPMapper on identical prepared inputs and one CPU model. We distinguish standard binary SLAP, called on the uncut graph, from an expanded protocol combining binary and weighted modes, both directions and every single-edge cut. The latter is our exploration wrapper, not default SLAP or its published benchmark. Its comparison with default SLAP changes both search scope and weight mode; it is not a sweep-only ablation. Both methods retain their original algorithms, and external evaluation applies the same chemical endpoint-equivalence rule. The saved seed study and paired study have separately recorded runs and source identities; their recovery counts are not pooled.

3.2 Catalogue and decoded-reference verification

We report two external coverage measurements. *Catalogue recovery* asks whether the reference belongs to the retained matching relation. *Decoded recovery* requires an actual injective mapping from a saved starting correspondence and its joint symmetry choices, in a retained branch and event window. A positive certificate is replayed and its event count independently checked. A catalogue certificate may suggest a candidate, but cannot alone establish decoded membership.

The decoded audits preserve exact annotated heavy-atom indices and unmatched-heavy-atom requirements. Unannotated hydrogens must admit a feasible completion. Golden catalogue verification additionally permits endpoint chemical symmetries preserving charge, isotope, hydrogen count and stereochemistry; the Golden catalogue and decoded scores therefore differ in both representation and reference-equivalence policy. EnzymeMap uses exact indices for both. Recovery means inclusion in either search direction, not top-ranked prediction accuracy. Every reaction remains in the denominator, and unresolved queries remain separate from absence.

3.3 Timing and branch-event analysis

Mapping time is the sum of matching and reduction; decoding is reported separately. CPU seconds sum successful worker computation over both directions and all cuts, including stage input loading and result construction. Publication, external verification, historical failed attempts and parallel elapsed duration have separate accounting. Timings use the saved successful passes and their documented recoveries, rather than a newly measured uninterrupted run of one final performance revision. Exact source and measurement boundaries are recorded in the repository.

To locate a recovered reference within the decoded event landscape, let $\mathcal{C}_{\text{GT}}$ contain the retained branch contexts that admit it inside the saved window. We measure

$$n_{\text{GT}} = \min_{c \in \mathcal{C}_{\text{GT}}} \min_{m \in c} E_c(m) - U, \tag{5}$$

where U is the overall detected count, minimized across both directions. Ground truth selects the qualifying contexts; their event minima are queried without fixing the reference assignments. Both terms use the same

saved all-atom decoder model. This measures where a GT-containing branch begins, not the physical event count of the reference itself. Missing references and unresolved offsets remain outside numeric bins.

We separately count the distinct heavy-atom mapping classes actually attainable at the detected minimum, taking the union across retained branches and both directions. Chemical endpoint equivalence identifies repeated representations, and hydrogen permutations do not multiply the choice count; each class must admit a full valid completion. The ranking score remains the saved all-atom model count. If a reaction has K choices and its reference class is present, a uniform selector succeeds with probability $1/K$; otherwise it succeeds with probability zero. This defines expected random-choice accuracy without a trained selector. Incomplete counts contribute bounds, not exact values. A GT-containing branch's minimum need not be attained by GT itself, so this measurement is separate from Equation (5).

4 Results

4.1 Reference coverage on Golden and EnzymeMap

All requested matching, reduction and decoding outputs are available: 3,702 directions in each Golden setting and 95,948 in EnzymeMap. Table 1 and figure 3 separates catalogue membership from the independently audited decoded window.

Table 1 Reaction-level reference coverage. Percentages use every reaction. Decoded columns use exact annotated indices in the saved model min+5 window. Golden catalogue membership also allows endpoint chemical symmetry; EnzymeMap catalogue membership uses exact indices. Unknown queries are not counted as misses.

Study	Reactions	Catalogue	Decoded	Absent	Unknown
Golden, 1 seed	1,851	1,834 (99.082%)	1,832 (98.974%)	16	3
Golden, 10 seeds	1,851	1,843 (99.568%)	1,841 (99.460%)	8	2
EnzymeMap, 1 seed	47,974	47,798 (99.633%)	47,796 (99.629%)	178	0

One-seed Golden decoding confirms 1,832/1,851 references (98.974%), with 16 absent and three unresolved. Ten seeds confirm 1,841 (99.460%), with eight absent and two unresolved. Ten references become newly confirmed; one previously positive reaction remains unresolved in the ten-seed audit, giving a net gain of nine confirmed reactions and no proven loss. Catalogue recovery is 1,834 and 1,843, respectively. One ten-seed directional catalogue query exhausted its allowance, but its opposite direction establishes that reaction's catalogue recovery.

EnzymeMap decoding confirms 47,796/47,974 references (99.629%), compared with 47,798 (99.633%) in the catalogue. The two additional catalogue references have 22-event witnesses outside their saved 14–19 windows; independent decoded queries confirm their exclusion. All 94,820 positive directional decoded certificates passed replay and count checks, with no invalid certificate. Forty-five reactions unresolved by bounded decoded queries are conclusively absent from the unchanged containing catalogue, establishing their absence from its decoded subset. Their 250 unresolved directional queries remain recorded as such. The combined reaction accounting is therefore 47,796 recovered, 178 absent and zero unresolved.

4.2 Separate mapping and decoding costs

Table 2 and figure 3 report compute CPU rather than parallel latency. One-seed Golden requires 1.31 CPU seconds per reaction for matching plus reduction and 1.88 for decoding. Ten seeds require 11.64 and 9.28 seconds, respectively. The corresponding EnzymeMap means are 4.41 and 8.75 seconds, with medians of 2.15 and 3.46 seconds. The median combined EnzymeMap cost is 5.80 seconds; medians of separate stages need not add.

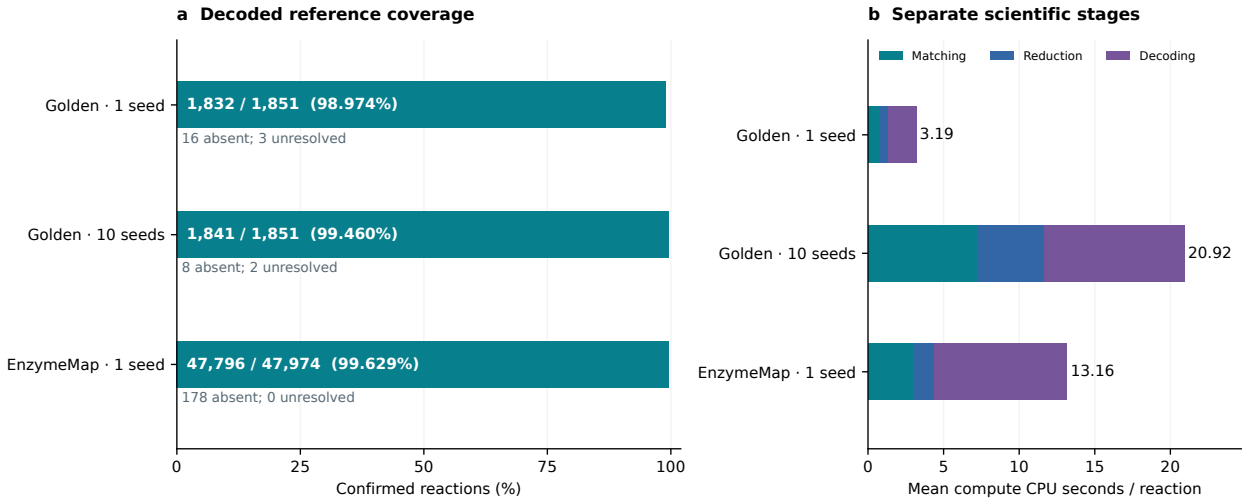


Figure 3 Reference coverage and computational cost. (a) Confirmed reference inclusion in the saved decoded min+5 window, with missing and unresolved reaction counts retained. This is inclusion among alternatives, not top-ranked accuracy. (b) Mean compute CPU per reaction, separated into matching, reduction and decoding. Both search directions contribute. Golden matching was reused for the corrected postprocessing passes; EnzymeMap combines original successful work with successful recoveries. External verification and publication are excluded.

Table 2 Mean compute CPU seconds per reaction. Both directions and the full cut sweep are included. Matching plus reduction is the mapping stage. Decoding is separate; publication and external verification are excluded. These are recorded completed-stage costs, not a hardware-normalized comparison with the archived methods.

Study	Matching	Reduction	Mapping	Decoding	Total
Golden, 1 seed	0.77	0.54	1.31	1.88	3.19
Golden, 10 seeds	7.27	4.37	11.64	9.28	20.92
EnzymeMap, 1 seed	3.02	1.39	4.41	8.75	13.16

The distributions have substantial tails. Golden's maximum directional decoding cost is 52.69 seconds with one seed and 316.46 seconds with ten; its corresponding medians are 0.476 and 1.964 seconds. EnzymeMap's combined per-reaction computation has a 95th percentile of 41.86 seconds and a 99th percentile of 128.58 seconds. Including publication and measured worker overhead raises EnzymeMap's mean from 13.16 to 14.60 CPU seconds per reaction. Historical failed attempts contribute a further approximately 27.93 sampled CPU hours; external verification and diagnostic work are excluded from these algorithm costs.

4.3 Seed count and computational scaling

Increasing Golden from one to ten seeds increases matching CPU by 9.42-fold, reduction by 8.09-fold and decoding by 4.95-fold. Matching therefore scales approximately linearly with seed count in the measured one-to-ten-seed comparison. Decoding grows less because the extra searches produce a reduced set of shared fragments and branch compositions rather than ten independent decoding jobs. These measurements describe the cost of additional exploration and its coverage gain; they are not a controlled isolation of individual acceleration mechanisms.

4.4 Paired coverage and speed comparison with SLAP

The paired study recovers 1,839/1,851 references with MAPPA (99.352%), 1,794 with expanded SLAP (96.921%), and 1,620 with standard binary SLAP across both directions (87.520%; Figure 4). Expanded SLAP has 52 confirmed misses and five unresolved reactions; standard bidirectional SLAP has 230 misses

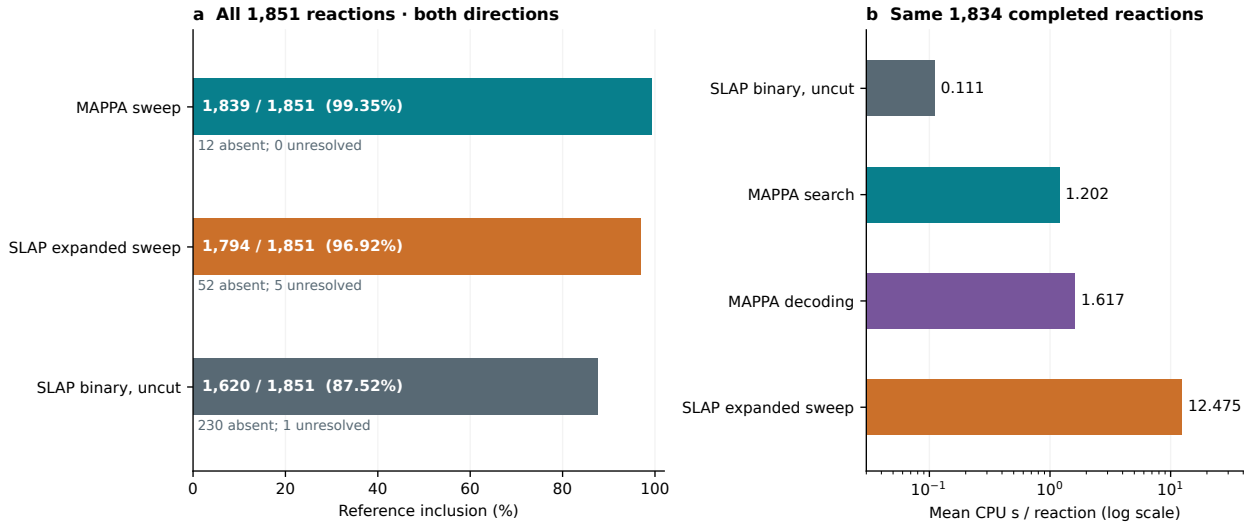


Figure 4 Coverage and cost depend on exploration scope. (a) Reference inclusion under identical chemical-equivalence evaluation over all 1,851 Golden reactions. Unknowns remain outside the recovered count. Standard SLAP is binary and uncut; expanded SLAP combines both weight modes and the cut sweep. All bars include both directions. (b) Mean compute CPU on the same 1,834 reactions completed by all protocols, with MAPPA decoding separate from search. The 17 excluded reactions remain in the coverage denominator. Timing is logarithmic and measures accumulated CPU, not parallel latency. The expanded wrapper raises SLAP coverage to approximately 97%, while increasing its computational cost.

and one unresolved. MAPPA has 12 misses and no unresolved catalogue queries. The independent comparator audit reconstructs its actual returned heavy mappings and repeats chemical-equivalence verification with a separate graph-matching implementation: all 3,683 completed expanded-SLAP direction verdicts agree.

Default forward-only SLAP recovers 1,591/1,851 (85.95%). This is close to its original paper's 86.9%, which also measures reference inclusion among alternatives but uses a 1,758-reaction Golden version (9). It is therefore an independent reproduction of similar behavior, not an exact replication of that dataset. Our archived first-returned default-SLAP score is 1,399/1,851 (75.58%). The retained alternatives tie at SLAP's lowest found cost; their output order is not a separate confidence ranking. In contrast, default single-answer interfaces such as RXNMapper and LocalMapper are evaluated directly on their selected prediction (Appendix C.1).

On the 1,834 reactions completed by all three protocols, mean search costs are 0.111 CPU seconds for standard binary SLAP, 1.202 for MAPPA, and 12.475 for expanded SLAP, summing both directions. MAPPA decoding adds 1.617 seconds. Thus MAPPA search is 10.38 times cheaper than the expanded SLAP sweep, but standard SLAP is 10.80 times cheaper than MAPPA search. The broader standard-SLAP/MAPPA pair contains 1,849 completed reactions and retains a slower standard-SLAP tail: their means are 0.487 and 1.259 seconds, a 2.58-fold difference. These are paired measurements on Intel Xeon Gold 6442Y processors, excluding loading, publication and external verification. Failed attempts remain separately accounted; excluded reactions are not treated as fast completions.

4.5 How many choices remain at the minimum?

The minimum-event rule produces a compact shortlist in the saved one-seed Golden study (Figure 5). Exact counts are established for 1,829 reactions; 22 remain incomplete under the counting allowance. Among the certified cases, 1,370 have one choice, 352 have two, 96 have three to five, and 11 have six or more. The certified-subset mean is 1.36 choices. Even allowing arbitrary larger counts for unfinished cases, the full-dataset median is one and its 95th percentile is three. We do not estimate the full-dataset mean from the completed subset.

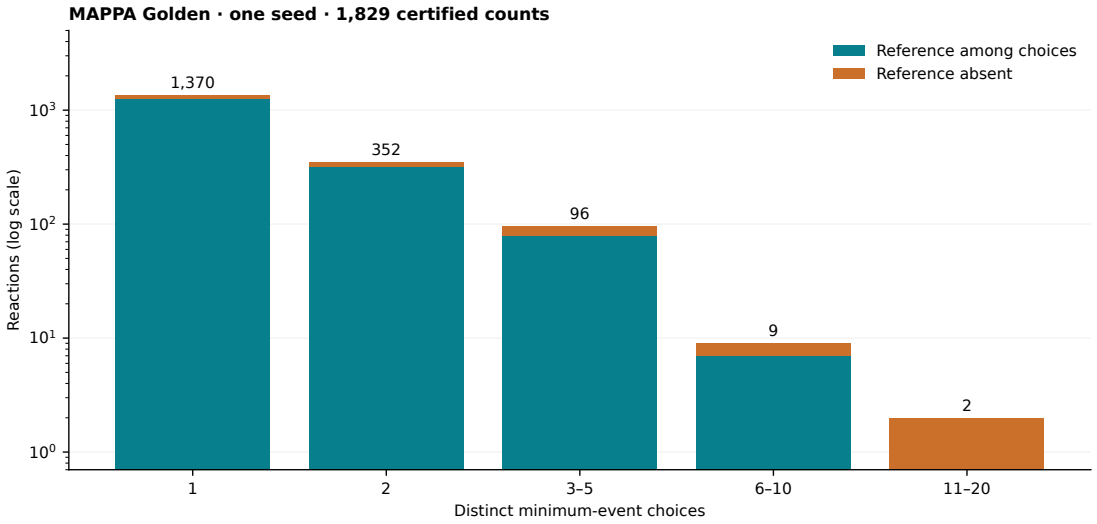


Figure 5 Distinct choices under minimum-event selection. Counts cover 1,829 of 1,851 one-seed Golden reactions with certified totals; 22 incomplete counts are excluded from the bars. Equivalent heavy-atom relabellings and hydrogen permutations count once. Colors indicate whether the reference is among the choices at the saved all-atom detected model minimum. The vertical axis is logarithmic. This is a count of attainable mapping classes, not branches or raw symmetry permutations.

The reference is confirmed at the detected minimum in 1,691 reactions, absent there in 158, and unresolved in two: minimum-set coverage is therefore 91.36–91.46%. Expected uniform-choice accuracy is 78.31–79.45%, with unfinished counts contributing explicit bounds. For example, Golden case 1694 has eight distinct minimum-event choices, including the reference, giving a random selector probability 1/8. These results quantify the decision left to a downstream selector. They do not establish top-1 superiority over another mapper, and a reference outside the minimum set would require selection from a wider event window.

4.6 Where reference-containing branches enter the event window

Figure 6 reports the preliminary distribution defined in Equation (5), with logarithmic reaction counts. Offsets are established for 45,737 reactions: 1,620 Golden and 44,085 EnzymeMap reactions have a GT-containing branch at the overall detected count; another 32 EnzymeMap reactions have offsets one through four. There are 3,891 pending offset checks, in addition to the separately reported absent and unresolved references.

Zero-offset certificates can establish the answer immediately, so the completed subset overrepresents zero. The higher-offset distribution remains incomplete and does not yet support full-dataset proportions. Both sides of the subtraction use the same decoder count, including hydrogen and the ordinary-boundary approximation. The offset measures the reported model landscape, not physical event-bin accuracy.

4.7 Chemical interpretation of retained alternatives

A correspondence describes net atom origins and bond changes without specifying intermediates or elementary steps. Figure 7 illustrates why retaining alternatives can matter. In Golden case 9, the annotated assignment places the alcohol oxygen in the product ring and the ketone oxygen in water. A second assignment exchanges those oxygen origins, while a third also changes two carbon assignments. The second has fewer events than the reference; the first and third have equal event totals but different correspondences. Neither a minimum event count nor a single total identifies the annotated transformation.

The figure uses selected archived witnesses as a chemical illustration. It is not an exhaustive output of the current decoder, and the alternatives are not validated reaction pathways. The historical coordinate/WBO

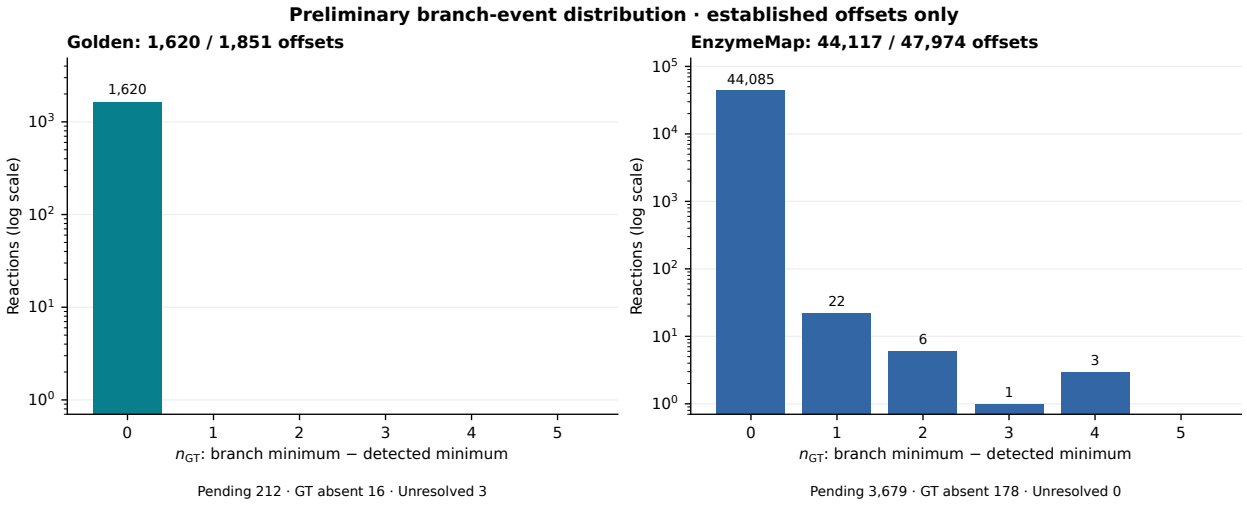


Figure 6 Preliminary location of GT-containing branches. The horizontal coordinate is the smallest event count of a retained GT-containing branch minus the overall detected count; the vertical axis is logarithmic. Ground truth selects the branch, rather than constraining the minimum witness within it. One-seed results use tolerance 2, cap 100 and saved min+5 responses. Pending checks and unresolved memberships are excluded from bars and labeled explicitly. Empty bins must not be interpreted as final zero frequencies.

study in Appendix C similarly examines event alternatives under unverified mappings; it is not a mapping-accuracy benchmark.

5 Discussion

MAPPA separates the existence question needed during growth from the compressed alternatives needed at saturation, and then separates correspondence discovery from event interpretation. Atom-set reduction gives repeated fragments one shared identity. The decoder reuses local symmetry-event responses while retaining correlated occupancy and the special dependence introduced by a swept bond. These module boundaries make search, representation and interpretation independently inspectable and reusable.

The benchmarks demonstrate high reference inclusion with complete saved mapping and decoding outputs, while preserving the distinction between catalogue and decoded coverage. More Golden seeds recover additional references at increased computation, with matching cost approximately proportional to seed count over the measured range. The paired SLAP comparison shows the trade-off directly: default SLAP is cheaper, whereas MAPPA explores a broader alternative set and costs less than the expanded SLAP sweep. Returning alternatives is shared by both methods; the comparison concerns their coverage, organization and cost.

Several limits remain scientifically important. Greedy trajectories, sampled seeds, single-edge cuts and hierarchical caps restrict discovery. Selecting one original fragment record per atom-set pair and one decoded starting orbit can omit alternatives present in discarded discovery records. Bounded improvement can retain an unproved minimum. None of these restrictions establishes exhaustive correspondence recovery.

The fixed ordinary-boundary model is also an approximation. In the EnzymeMap audit, 2,801 positive directional witnesses have different model and physical whole-reaction event counts; the ten-seed Golden audit records 32 such witnesses. Both counts are retained. Model-window membership is therefore not physical event-bin accuracy, and a low model count alone does not identify the annotated or most plausible chemistry. The branch-offset figure assesses organization within that model, not a replacement accuracy measure.

Retained alternatives can support reaction-path preparation: atom-index assignment affects endpoint interpolation (3), and different permitted correspondences could supply starting structures for nudged elastic band calculations (8). A separate chemical selector could turn the same catalogue into a single-answer pre-

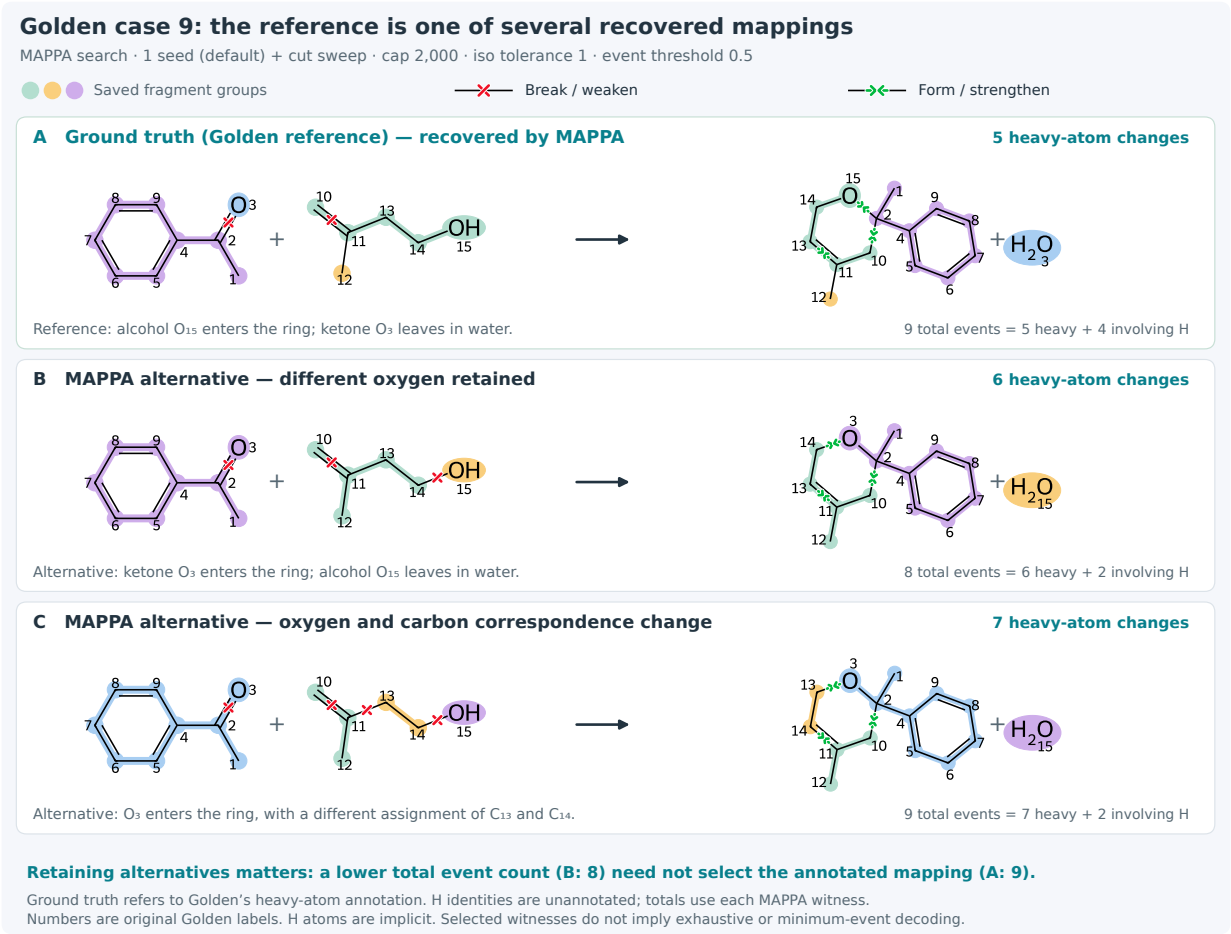


Figure 7 A reference correspondence and two alternatives for Golden case 9. A matches all 15 annotated heavy-atom pairs. B exchanges the oxygen origins; C also exchanges C_{13}/C_{14}. Colors identify matched fragments. Black lines show endpoint bonds, red × marks show cleavage or weakening, and green inward arrows show formation or strengthening. H atoms are implicit in the drawing but included in event totals; each witness minimizes H-involving changes for its fixed heavy-atom correspondence. "Ground truth" denotes the dataset annotation. These archived witnesses were obtained without reference labels using one-seed sweep search, cap 2,000 and $\tau_{\text{iso}} = 1$; events use threshold 0.5.

diction, using minimum event count as a surrogate and reaction-specific preferences to resolve ties or retain a higher-event candidate. These are downstream opportunities, not demonstrated improvements in ranking or transition-state prediction. The current Golden and EnzymeMap studies also do not validate continuously valued WBO inputs or every stereochemical attribute. The archived comparator, seed/cap, direction and coordinate analyses remain in the supplement with their original protocols; only the separately paired current comparison supports its stated speed ratios.

Code and data

Code and reproduction instructions are available at MAPPA source repository, branch `main`. The repository guide `manuscript/REPRODUCIBILITY.md` links full benchmark configurations, software versions, per-case results, and figure-generation scripts. Golden and EnzymeMap are distributed by their original authors (10; 7).

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References

[1] Shuan Chen, Sunggi An, Ramil Babazade, and Yousung Jung. Precise atom-to-atom mapping for organic reactions via human-in-the-loop machine learning. *Nature Communications*, 15:2250, 2024. doi: 10.1038/s41467-024-46364-y.

[2] Chython contributors. Chython: molecular and reaction processing toolkit. https://github.com/chython/chython. Evaluated version 2.18 with chython-rxnmap 2.0; model hash archived.

[3] Deb Sankar De, Marco Krummenacher, Bastian Schaefer, and Stefan Goedecker. Finding reaction pathways with optimal atomic index mappings. *Physical Review Letters*, 123:206102, 2019. doi: 10.1103/PhysRevLett.123.206102.

[4] Leonardo de Moura and Nikolaj Bjørner. Z3: An efficient SMT solver. In *Tools and Algorithms for the Construction and Analysis of Systems*, pages 337–340. Springer, 2008. doi: 10.1007/978-3-540-78800-3_24.

[5] EPAM Systems. Indigo cheminformatics toolkit. https://github.com/epam/Indigo. Evaluated version 1.46.0.

[6] Thomas D. Goddard, Conrad C. Huang, Elaine C. Meng, Eric F. Pettersen, Gregory S. Couch, John H. Morris, and Thomas E. Ferrin. UCSF ChimeraX: Meeting modern challenges in visualization and analysis. *Protein Science*, 27:14–25, 2018. doi: 10.1002/pro.3235.

[7] Esther Heid, Daniel Probst, William H. Green, and Georg K. H. Madsen. EnzymeMap: curation, validation and data-driven prediction of enzymatic reactions. *Chemical Science*, 2023. doi: 10.1039/D3SC02048G. https://doi.org/10.1039/D3SC02048G.

[8] Graeme Henkelman, Blas P. Uberuaga, and Hannes Jónsson. A climbing image nudged elastic band method for finding saddle points and minimum energy paths. *The Journal of Chemical Physics*, 113(22):9901–9904, 2000. doi: 10.1063/1.1329672.

[9] Shin-ichi Koda and Shinji Saito. General and scalable atom-to-atom mapping via Weisfeiler–Lehman-like approximate graph matching. *ChemRxiv*, 2025. doi: 10.26434/chemrxiv-2025-hthwn. Preprint, version 1, 25 November 2025.

[10] Arkadii Lin, Natalia Dyubankova, Timur I. Madzhidov, Ramil I. Nugmanov, Jonas Verhoeven, Timur R. Gimadiev, Valentina A. Afonina, Zarina Ibragimova, Assima Rakhimbekova, Pavel Sidorov, Andrei Gedich, Rail Suleymanov, Ravil Mukhametgaleev, Joerg Wegner, Hugo Ceulemans, and Alexandre Varnek. Atom-to-atom mapping: A benchmarking study of popular mapping algorithms and consensus strategies. *Molecular Informatics*, 41(4):2100138, 2022. doi: 10.1002/minf.202100138.

[11] Brendan D. McKay and Adolfo Piperno. Practical graph isomorphism, II. *Journal of Symbolic Computation*, 60: 94–112, 2014. doi: 10.1016/j.jsc.2013.09.003.

[12] Eric F. Pettersen, Thomas D. Goddard, Conrad C. Huang, et al. UCSF ChimeraX: Structure visualization for researchers, educators, and developers. *Protein Science*, 30:70–82, 2021. doi: 10.1002/pro.3943.

[13] S. A. Rahman, G. Torrance, L. Baldacci, S. Martínez Cuesta, F. Fenninger, N. Gopal, S. Choudhary, J. W. May, G. L. Holliday, C. Steinbeck, and J. M. Thornton. Reaction decoder tool (RDT): extracting features from chemical reactions. *Bioinformatics*, 32(13):2065–2066, 2016. doi: 10.1093/bioinformatics/btw096.

[14] Philippe Schwaller, Benjamin Hoover, Jean-Louis Reymond, Hendrik Strobelt, and Teodoro Laino. Extraction of organic chemistry grammar from unsupervised learning of chemical reactions. *Science Advances*, 7(15):eabe4166, 2021. doi: 10.1126/sciadv.abe4166.

A Technical algorithm details

A.1 Exact existence queries during growth

An existence query contains the proposed source atoms, their accepted preservation edges and completed-fragment locks. Element-compatible destination domains exclude targets already occupied by other fragments. Each accepted edge supplies a binary support relation satisfying target bond presence and the discovery tolerance. Source nonedges and unselected molecular edges do not add constraints.

Propagation removes occupied singleton destinations and destinations without support in a neighbor's domain. An empty domain or incompatible singletons establish infeasibility. A bipartite distinct-target check can reject domains that cannot accommodate an injective assignment. These are necessary conditions; surviving ambiguous cases require search. The solver chooses a constrained atom, propagates its tentative destination and stops at the first complete witness. Proving absence can require exploring every remaining feasible choice. Positive and negative answers use the same constraints.

Direct witness extension and indexed fragment reuse precede this query. They establish existence only. Neither supplies a different continuation policy nor converts a timeout into absence. Completed-fragment assignments remain fixed while assignments internal to the current growing fragment may change.

A.2 Completion, local symmetry and global placement

Completion searches the finished fragment constraints for distinct target atom-set placements. It excludes already represented sets or certified global placement orbits within the search. It does not enumerate all internal assignments before deduplicating them. Target sites that cease to be globally equivalent remain separate placements, even when their local matched subgraphs share symmetry.

For each local fragment, nauty receives an element-colored binary graph with individually fixed anchors. Its compact generators retain correlated moves of rings, arms and their attached atoms. Continuous weights do not label this graph; they remain in discovery and event predicates. Permitted local rearrangements are compositions of these binary generators. An accepted discovery mapping can be followed by a source and a target local action without reimposing discovery tolerance on the rearranged result. The event response evaluates the resulting original-weight changes.

The global action used to compress later continuations is more restrictive. It must preserve the full target weight matrix, active connectivity, current locks and every source-to-target compatibility predicate relevant to subsequent growth. Such an action moves the whole later continuation, preserving occupied targets and its matching problem. A local fragment action cannot be substituted for this global equivalence proof. Saved placement operations retain their order; unrestricted closure of an ordered sequence would permit combinations not represented by the search.

A.3 Hierarchical caps and provenance

Each retained placement creates a child continuation. After a completed step, count surviving continuations bottom-up. At an over-limit node, discard its largest child subtree, recount and repeat; ties retain the earlier decision. An ancestor then sees the reduced count. The completed-fragment limit separately rejects a placement expansion exceeding its allowance. Neither operation counts internal symmetry assignments as separate branches, and both retain explicit cap evidence.

A shared search state may have several incoming histories. The retained graph keeps parent links and terminal references rather than unfolding every path. States and payloads with no surviving descendant are released. Returned histories and cut provenance remain available for inspection, but are not independent fragment-decoding jobs. A branch cap bounds retained choices, not every temporary allocation or total memory.

A.4 Fragment identities and composition reduction

Reduction reuses the fragment inventory in each search graph and unifies it across seeds and cuts. A completed fragment's key is its source/target atom-set pair within a fixed endpoint problem and policy. Keep the first original record intact. Other starting witnesses, growth routes, domains or anchors do not create additional identities and are not unioned into broader permissions. Raw archives retain the original decisions; the reduced catalogue retains the selected record and compact provenance. When reducing an explicitly selected path subset, the representative must come from that subset.

Native projection converts retained states into shared fragment identifiers. Composition keys use those identifiers together with the ordered placement context. Independent decisions can commute; overlapping local or global choices retain their order. Equal keys share one family record and aggregate cut/partition provenance. Reduction uses structural equality, not symmetry solving or event queries. The atom-set key must not be confused with a permission to map its atoms arbitrarily.

A.5 Shared event core and bounded starting-map improvement

Each unique fragment is prepared once. For a selected starting correspondence, the shared core stores binary symmetry generators, fixed event contributions and sparse variable original-weight terms. It separates operation components only when no variable event or requested endpoint joins them. A generator moving two arms couples those moves even if each arm also has its own independent internal exchanges. Event-preserving operations can share computation only when they also preserve all endpoints relevant to the query. No complete group or Cartesian table of mappings is stored.

Starting-map improvement first compares the saved witness with lower bounds. One bound matches pair-weight multisets within element-pair types while relaxing atom consistency. Another matches incident bond assignments and then atoms injectively; each event touches two atoms, so half the incident cost, rounded up, is a lower bound. The stronger bound is retained. These relaxations can rule out improvement but need not be attainable.

Optimize the selected fragment's binary symmetries. Search additional discovery correspondences only when its recorded growth connectivity differs from final internal connectivity on either endpoint and the current witness remains unsettled. New starting correspondences must satisfy the saved discovery domains, anchors and accepted edges; subsequent binary symmetry choices are not rejected by those discovery constraints. Preparation, correspondence improvement and symmetry optimization share a five-second per-fragment deadline. Attaining a valid bound stops work immediately. Otherwise retain an actual best witness and incomplete status. An orbit minimum does not certify a minimum over discarded starting orbits.

Branch assembly binds local response operations to their shared branch choices. Final partition pieces own disjoint internal contributions. Weight-preserving global placements move endpoints while retaining those local costs. Ordinary boundaries remain fixed under the declared model. The sweep correction follows jointly reachable endpoint pairs through the saved action order, preserving dependence on variable internal events. When internal costs are constant, this is an endpoint-pair calculation, not a complete symmetry-assignment enumeration.

For a requested event interval, the saved response searches its retained choices and returns a concrete mapping with its ordered choices, absence after complete exclusion, or unknown under a limit. Event counts and mappings can be replayed independently. Preparing a queryable response does not assert that every count in its numerical interval is attainable. Physical whole-reaction event counts can differ from the independent-boundary model and are recorded separately during external audits.

A.6 Indexed fragment reuse and execution

The reaction-local pool stores one original fragment per atom-set pair. A restricted saved witness proves a growth proposal when it covers the requested source atoms, uses available target atoms, respects locks and supports every required edge at the current tolerance. Atom-set containment alone is insufficient when occupancy or edge requirements have changed. An inverted index intersects atom, edge-support and fixed-

assignment sets to find a valid proof. A miss invokes the ordinary existence query; no negative cache is used.

Growth continues with its own random proposals. Cut/trial streams are distinct and reproducible, and completion chooses its representatives independently of the temporary existence witness. The pool therefore does not replay a previous trajectory. Its accounted retention limit bounds cached records and index storage, not total stage memory. Returned search graphs expose their unique-fragment inventories directly to reduction.

Persistent execution shares compatible reaction snapshots at task boundaries. A permanent first snapshot and replaceable latest snapshot support concurrent readers; a validated first snapshot can be used when the latest stored version cannot be loaded. This is recovery of reusable data, not an alternative scientific algorithm. A single publication owner atomically replaces complete versions, and stale publishers cannot overwrite newer ones. Stored snapshot count is bounded per reaction rather than per cut or seed.

Separate stage APIs return typed matching, catalogue and decoding records. In-memory and aggregated file-backed transport preserve shared immutable arrays and fragment relations. Worker pools schedule those operations without changing their scope. For the reported batch protocol, a global matching/reduction barrier precedes decoding. Recovery retains completed outputs and consumed budgets; resource errors remain distinguishable from mapping absence. The main manuscript reports successful compute costs separately from failed attempts and external evaluation.

B Reproducibility and measurement scope

The repository's `manuscript/REPRODUCIBILITY.md` links source pins, immutable input identities, benchmark records and the evidence used to generate tables and figures. The public Python API documents direct matching, reduction, decoding and conditional queries. Figure building runs no searches. Publication records omit execution-site identifiers.

The main-text Golden measurements combine the saved one- and ten-seed matching passes with corrected completed reduction and decoding passes. They are not new uninterrupted end-to-end measurements. The matching passes predate indexed containment; the paper treats their seed scaling as an empirical workload comparison, not a cache ablation. EnzymeMap combines original successful tasks and successful recoveries with unchanged scientific settings. Performance corrections preserved the mapping relation and event model. CPU totals include computation, stage loading and record construction; publication, external audits and sampled earlier failures are separate. CPU models were not retained in the exported full-campaign summaries, so no hardware-normalized ranking follows.

Every accepted decoded-reference witness is checked for replay, injectivity, annotated assignments and an independently counted model total inside the saved window. Its physical total is stored separately. The Golden decoded audit uses exact indices; its catalogue audit permits endpoint chemical equivalence. EnzymeMap uses exact indices for both. A conclusive absence from the unchanged containing catalogue implies absence from the decoded subset; that inference is recorded separately from a direct negative decoded query. Bounded unknown queries are never relabeled as direct negatives.

The branch-offset snapshot contains one row per reaction. Its 45,737 exact offsets, 3,891 pending offsets, 194 absent references and three unresolved memberships sum to 49,825. Easy zero-offset certificates are resolved first, so the partial histogram is not an unbiased estimate of the final distribution. Both compared counts use the saved all-atom decoder model. No heavy-only reference count enters the subtraction.

The supplementary September studies below retain their original settings, evaluators and comparison cohorts. Their archived paired search timings use an Apple M4 Pro CPU; the cap comparison uses an AMD EPYC 9J14. Search, publication and event-query boundaries differ across those studies and are stated with each table. They are not pooled with the current main-text timings. The WBO-generation method for the archived coordinate inputs is not recorded in the available metadata, so regeneration of those matrices from molecular structures is not established.

Table A1 Archived evaluation configurations. Both use $b = 0.2$, $\tau_{iso} = 1$, explicit H, and uncut plus single-edge searches. One seed per cut and the sweep define the default setting. Archived Golden measurements cover sweep search; no-sweep controls retain only the uncut graph. A seed setting counts source-atom orderings per cut.

Setting	Golden	Coordinate/WBO
Reactions	1,851	140
Search directions	Bidirectional	Reactant to product
Seed orderings	1 (default), 2, 3, 10	1 (default)
Branch cap	100 (default)	100 (default), 2,000
Evaluation	Reference-family recovery	Signed event-class coverage

C Archived September evaluations

The following studies used their recorded September implementations. They retain the original numerical results and comparison scopes; they are not reruns of the current algorithm or direct performance comparisons with Section 3.1.

C.1 Archived datasets and comparator protocol

Golden inputs use formal bond orders and explicit H, with reference labels removed before search. Recovery requires the annotated heavy-atom correspondence, including unmatched atoms, modulo endpoint chemical symmetries preserving charge, isotope, H count, and stereochemistry. These chemical colors define reference equivalence; they do not imply that element/WBO matching itself enforces every chemical attribute. Failures and unresolved cases remain in the denominator. This measures reference inclusion among returned alternatives, not their ranking.

LocalMapper provides the prior accuracy-SOTA baseline (1). Its published Golden accuracy was 89.8% on 1,758 retained reactions with manual calibration. SLAP reported 86.9% on that dataset version, counting reference inclusion among its candidates (9). These literature scores are separate from our strict re-evaluation. We additionally apply the cut sweep introduced here to SLAP; "SLAP sweep" denotes this extension, not a published SLAP score. It is our strongest evaluated prior-method baseline for alternative recovery. Released RXNMapper, Chython, Indigo, and RDT provide further comparisons (14; 2; 5; 13).

The coordinate collection contains 18–137 explicit atoms per endpoint, including organometallic and multi-component structures. It was inspected during development and has no verified atom mappings: it is not a mapping-accuracy benchmark. The archived study compared the lowest event count found and the patterns achieving that count. MAPPA used supplied WBO matrices; SLAP used XYZ-derived binary graphs. Returned SLAP heavy-atom assignments received hydrogen refinement and were scored on the same WBO matrices as MAPPA. The comparison covers those saved witnesses, not all possible SLAP hydrogen assignments or a fresh current-implementation evaluation.

Released comparators received the same label-free Golden inputs, without prediction repair or reference-guided ranking. Invalid endpoint chemistry, duplicate atom labels and other failures remained in the denominator. Binary SLAP was retained as the stronger default configuration; bidirectional comparisons used both binary and weighted modes. SLAP returns multiple candidates, whereas the other comparators return at most one explicit bijection. These results describe the evaluated interfaces and scoring rule, including its difference from RDT's chemical-change criterion. The repository records released models, versions, adapter settings and failure frequencies.

C.2 Archived evaluation configurations

Table A1 lists the archived settings. Search did not use reference or comparator correspondences. Full configurations, implementation versions, and reproduction scripts are provided in the repository (Appendix B).

Table A2 Golden reference recovery and runtime in the archived evaluation. Recovery uses all 1,851 primary-evaluation records. MAPPA retains compressed families; SLAP returns explicit candidates; other methods return one bijection over their matched atoms. MAPPA uses cap 100 and bidirectional search. Bidirectional SLAP combines both directions and both binary/weighted modes; the default SLAP row uses binary only. Times are mean CPU seconds per reaction, excluding separate bond-event decoding. Unmarked times use the same 1,403 completed reactions on an AMD EPYC 9J14. *a*: archived completed calls (1,718–1,851 per method), CPU model unrecorded; these times are not directly comparable with unmarked times. †: LocalMapper, prior accuracy SOTA (2024). ‡: SLAP with our sweep, strongest evaluated prior-method recovery baseline.

Method	Search setting	Output	Reference recovered	Mean CPU s
MAPPA	1 seed, sweep (default)	Families	1,834 (99.08%)	0.963
MAPPA	2 seeds, sweep	Families	1,834 (99.08%)	1.619
MAPPA	3 seeds, sweep	Families	1,837 (99.24%)	2.265
MAPPA	10 seeds, sweep	Families	1,840 (99.41%)	6.993
MAPPA	1 seed, no sweep	Families	1,489 (80.44%)	0.040
SLAP, bidirectional	No sweep	Candidates	1,661 (89.74%)	0.105
SLAP, bidirectional‡	Sweep	Candidates	1,796 (97.03%)	2.793
SLAP, binary	Default, no sweep	Candidates	1,591 (85.95%)	0.075[a]
RXNMapper	Default, no sweep	One bijection	1,547 (83.58%)	0.056[a]
LocalMapper†	Default, no sweep	One bijection	1,605 (86.71%)	0.294[a]
Chython	Default, no sweep	One bijection	1,561 (84.33%)	0.612[a]
Indigo	Default, no sweep	One bijection	692 (37.39%)	0.184[a]
RDT	Default, no sweep	One bijection	1,030 (55.65%)	1.021[a]

C.3 Archived Golden reference recovery and search cost

In the archived campaign, the one-seed MAPPA configuration recovered 99.08% of Golden references, compared with 97.03% for SLAP sweep (Table A2 and figure A1). These scores belong to that campaign, separate from Section 3.1. The paired comparison includes 43 references recovered only by MAPPA and 5 only by SLAP. LocalMapper is the strongest evaluated method returning one bijection. Additional MAPPA seeds yielded modest recovery gains at increasing search cost; two seeds did not improve recovery over one.

The paired archived controls attribute 345 additional reference recoveries to the MAPPA sweep and 135 to the SLAP sweep. Without the sweep, one-seed MAPPA had lower coverage than bidirectional SLAP. This supports varying the growth constraints to expose alternatives excluded by greedy fragment extension (Section 2.6); symmetry expansion alone cannot restore discarded placements. These differences are not measurements of a current no-sweep control.

Smaller-first grows fragments from the reaction side with fewer explicit atoms (including H) and matches into the other side; larger-first reverses these roles. In the archived comparison, combining both directions added 31 recoveries over smaller-first alone, demonstrating complementary alternatives (Table A3).

In these archived measurements, one-seed MAPPA had a lower mean and median search cost than SLAP sweep, but a higher 95th percentile. Two seeds approximately doubled the mean cost without improving recovery. These Golden timings cover search, excluding reference verification and separate bond-event decoding. The two timing tables use different experiments and should be compared within their stated CPU and reaction sets; neither establishes a current-implementation speed ranking.

In the archived cap study, increasing the branch cap from 100 to 2,000 yielded no additional swept recoveries among cases resolved at both caps, while increasing cost and unresolved outcomes at higher seed counts (Table A8). The uncut search gained 19 confirmed recoveries. Thus, in that study a larger branch budget did not substitute for changing the growth conditions.

Table A3 Archived recovery and runtime by search orientation. All rows use the sweep. Recovery uses all 1,851 Golden records; CPU statistics are seconds per reaction over the same 1,821 completed reactions (Apple M4 Pro CPU). Smaller-first grows from the side with fewer explicit atoms, including H; larger-first reverses the source and target. Equal-size ties use stored order and its reverse. Bidirectional output combines both searches, and its CPU cost is their sum. SLAP combines binary and weighted modes. CPU/SLAP compares means within each orientation. Timing scopes are specified in Appendix B.

Method	Orientation	Recovered	Mean	Median	95th pct.	CPU/SLAP
MAPPA, 1 seed (default)	Smaller-first	1,803 (97.41%)	1.141	0.263	5.289	0.34
	Larger-first	1,779 (96.11%)	1.377	0.348	6.401	0.92
	Bidirectional	1,834 (99.08%)	2.518	0.719	12.913	0.52
MAPPA, 2 seeds	Smaller-first	1,815 (98.06%)	2.232	0.416	10.635	0.67
	Larger-first	1,805 (97.51%)	2.721	0.578	14.064	1.83
	Bidirectional	1,834 (99.08%)	4.953	1.250	26.350	1.03
SLAP sweep (prior baseline)	Smaller-first	1,775 (95.89%)	3.329	0.422	4.752	1.00
	Larger-first	1,770 (95.62%)	1.490	0.455	4.568	1.00
	Bidirectional	1,796 (97.03%)	4.819	0.881	11.048	1.00

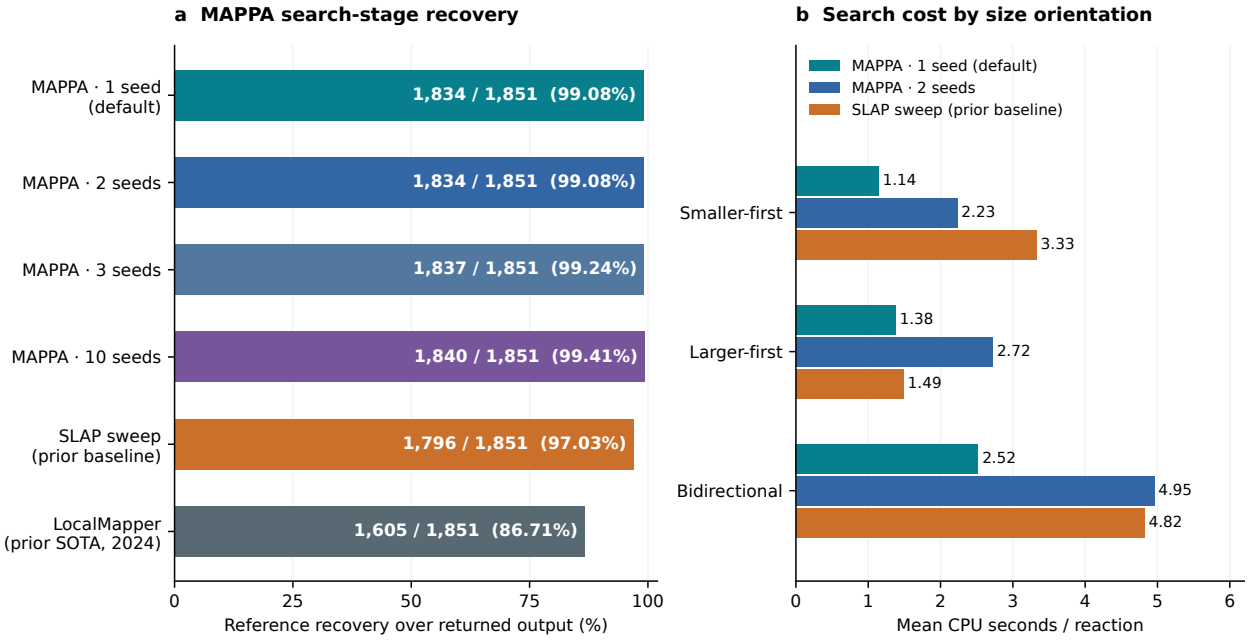


Figure A1 Archived Golden reference recovery and search cost. (a) Reference recovery over all 1,851 reactions. MAPPA and SLAP use our cut sweep; LocalMapper is the prior accuracy-SOTA baseline and returns one bijection. All scores use the archived evaluator. (b) Mean search CPU over the same 1,821 completed reactions, by orientation. Bond-event decoding is excluded. Full comparisons and timing scopes are in Tables A2 and A3.

C.4 Archived bond-event comparison on 140 coordinate reactions

The archived coordinate study used one-seed MAPPA sweep search with a branch cap of 2,000 and returned complete mappings for all 140 reactions. At cap 100, cases 123 and 125 returned no complete mapping. The 2,000-branch configuration therefore supplies this comparison, while 100 remains the library default. These completion counts and the event-window results below have not been remeasured with the current implementation.

The archived MAPPA output matched or lowered the minimum event count found by both native SLAP and SLAP sweep in every reaction (Table A4 and figure A2). This does not imply recovery of every alternative: at equal minima, six SLAP-sweep patterns remained absent across four reactions. An equal count can describe different changed bonds, and a lower count alone does not establish a more plausible transformation.

Table A4 Archived minimum bond-event comparison on 140 reactions. MAPPA uses one-seed sweep search at cap 2,000. The first three columns count reactions by MAPPA's minimum event count relative to the comparator. The final column gives comparator minimum-event patterns recovered by MAPPA among reactions with equal minima. Counts refer to returned candidates, not proven global optima.

Compared with	MAPPA fewer	Equal	MAPPA more	Patterns at equal minima
SLAP sweep	4	136	0	158/164
Native SLAP	15	125	0	138/140

To compare alternatives above MAPPA's own minima, the archived study queried through one event above the higher of the two SLAP minima for each reaction. MAPPA retained 162 of 168 SLAP-sweep minimum-event patterns and 151 of 160 native-SLAP patterns within these windows. The output contains 300 distinct patterns, including 166 at MAPPA's own minima. Every saved family was exhausted or certified within its window; this establishes completeness for those retained families, not exhaustive structural search.

Table A5 Archived stage CPU time per reaction on the 140 coordinate reactions. Measurements use an Apple M4 Pro CPU. MAPPA uses one-seed sweep search at cap 2,000. These stage costs exclude MAPPA decoding and do not establish an end-to-end speed ranking. SLAP refinement covers its returned witnesses. Timing boundaries are specified in Appendix B.

Recorded stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.047	0.314	3.481
SLAP sweep mapping	1.337	0.546	5.919
SLAP H refinement / initial scoring	0.053	0.011	0.207

The stage timings in Table A5 cover different outputs: MAPPA search retains families for subsequent decoding, while SLAP refinement operates on returned witnesses. They therefore do not establish an end-to-end speed ranking.

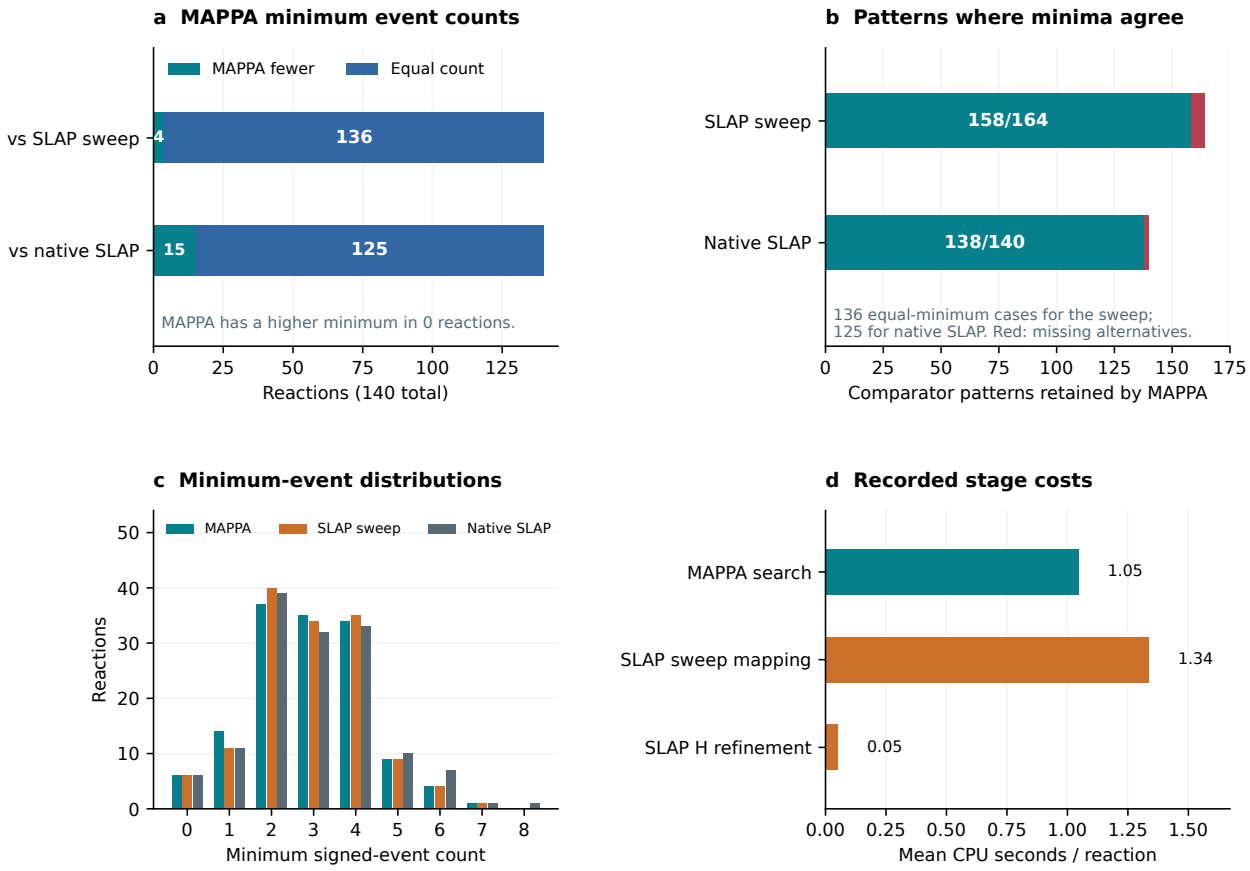


Figure A2 Archived bond-event comparison on 140 coordinate reactions. MAPPA uses one seed, the sweep, and cap 2,000. (a) Reactions with fewer or equal MAPPA events than each SLAP baseline. (b) Comparator minimum-event patterns recovered by MAPPA at equal minima. (c) Minimum-event count distributions. (d) Mean CPU for search and SLAP refinement, excluding MAPPA decoding (Table A5). Atom mappings are unverified; this compares bond events, not mapping accuracy.

C.5 Archived event-window query scope

For each coordinate reaction, the archived event-window limit was one event above the larger of the native-SLAP and SLAP-sweep minima. All 120,052 retained families were exhausted within these limits, without a class-count cap. Higher-event patterns are outside this completeness claim. An event class records the signed atom-pair changes, modulo source permutations preserving every thresholded bond-change test against the opposite endpoint. Each class has a concrete mapping witness. This explicit whole-reaction query is separate from ordinary decoding into shared local fragment responses.

Table A6 Archived complete event-window queries on all 140 reactions. Every retained family is certified within the listed per-case event limit.

Maximum events	1	2	3	4	5	6	7	8	9
Reactions (all complete)	6	11	39	32	33	10	7	1	1

The query used event invariance to certify constant families and encoded remaining choices in Z3 (4). A family was certified by exhaustion, a valid lower bound excluding the window, or unsatisfiability after excluding all discovered classes. An exhausted resource budget remained incomplete. The explicit query adapter can simplify adjacent nested action groups by exact generator-membership checks: if $H \leq G$, then $HG = GH =$

G. This adapter simplification is separate from ordinary catalogue reduction in Appendix A.4. Completeness within an event window concerns the saved families, not exhaustive structural search.

C.6 Archived Golden nonrecoveries

Table A7 The eleven archived ten-seed Golden nonrecoveries. Case indices are zero-based. Retained pair counts cover every terminal in both directions and every saved cut at ten seed orderings and cap 100. The four cardinality mismatches concern exact reference relations, including unmatched atoms. This is not the nonrecovery set of the current one-seed validation.

Case	Reference pairs	Retained pairs	Observed mismatch
7	29	30	One extra matched pair
19	23	23	Atom pairing excluded by orbit bounds
590	31	31	Atom pairing excluded by orbit bounds
871	20	20	Atom pairing excluded by orbit bounds
986	16	16	Atom pairing excluded by orbit bounds
1228	12	12	Atom pairing excluded by orbit bounds
1285	14	14	Joint C/O correspondence excluded
1358	36	37	One extra matched pair
1377	22	23	One extra matched pair
1475	18	19	One extra matched pair
1553	20	20	Atom pairing excluded by orbit bounds

Independent reconstruction agrees with the supplied references in all eleven cases. Four require a different number of matched pairs; seven require correspondences absent from the retained families. All have at least one capped direction, but this does not identify the cap as the cause. The archived uncapped traces below demonstrate greedy-growth exclusions in two cases. Reference-directed feasibility checks are diagnostic and do not contribute to benchmark recovery. Per-case certificates and traces are linked in the repository.

Two archived traces show how greedy extension excluded a representable reference. One reference exchanges adjacent labels in an otherwise unchanged pyridine ring; another exchanges the carbonyl-oxygen attachments of oxalic acid across its retained carbons. Growth preserved bonds that these correspondences require separating. Uncapped trajectories still missed both references, while reference-constrained fragment choices established their representability. In the oxalic-acid example, a reference-selected three-edge relaxation also exposed the reference to matching; the single-edge sweep did not contain that cut set. These are feasibility diagnostics, not additional benchmark recoveries or traces of the current matcher. For the other five correspondence misses, the audit did not isolate cap truncation from unsampled growth choices. They motivate selective alternatives to successful growth extensions and explicit choices about leaving atoms unmatched, rather than treating larger caps or more symmetry expansion as sufficient remedies.

Only one of the eleven misses is recovered by any tested comparator configuration, including the SLAP sweep. The shared failures warrant inspection of the supplied correspondences as well as search improvements; they do not establish annotation errors. All eleven remain in the reported denominator. Independent evaluation on annotated coordinate reactions is needed to assess generalization beyond the development collection.

C.7 Archived branch-cap and CPU-time ablation on Golden

The archived study ran the twelve configurations in Table A8 on all 1,851 Golden reactions using an AMD EPYC 9J14 CPU. Both search directions are unioned. SLAP unions its binary and weighted modes. Search and reference verification have separate fixed resource budgets. The recorded cap-100 recovery counts reproduce the archived primary evaluation. At cap 2,000, unresolved counts increase with seed count. Among reactions resolved at both caps, no swept configuration gains or loses a reference recovery.

The ten-seed cap-2,000 run leaves five references recovered at cap 100 unresolved, accounting for its lower confirmed count. The bounded SLAP sweep confirms 1,794 references here, compared with 1,796 in the archived

Table A8 Archived Golden branch-cap ablation with paired SLAP baselines. Recovery counts confirmed reference inclusion over all 1,851 records; unknown cases remain in the denominator. CPU values are seconds per reaction. Paired CPU uses the same 1,403 reactions whose searches finish under every configuration. Spent CPU averages recorded search cost over all 1,851 selected attempts, including interrupted calls; it is not completed-search latency. Both timings exclude reference verification and initial imports. Superseded attempts are excluded. Cap 100 with a one-seed sweep is the default setting.

Method	Cap	Seeds	Recovered (%)	Unknown	Paired CPU	Spent CPU
MAPPA uncut	100	1	1,489 (80.44)	0	0.040	0.085
MAPPA sweep (default)	100	1	1,834 (99.08)	0	0.963	3.873
MAPPA sweep	100	2	1,834 (99.08)	0	1.619	7.260
MAPPA sweep	100	3	1,837 (99.24)	0	2.265	10.556
MAPPA sweep	100	10	1,840 (99.41)	0	6.993	33.211
MAPPA uncut	2,000	1	1,508 (81.47)	5	0.086	0.517
MAPPA sweep	2,000	1	1,834 (99.08)	2	4.317	30.055
MAPPA sweep	2,000	2	1,834 (99.08)	4	8.283	48.549
MAPPA sweep	2,000	3	1,837 (99.24)	6	12.201	61.890
MAPPA sweep	2,000	10	1,835 (99.14)	12	43.843	113.777
SLAP uncut	–	–	1,661 (89.74)	1	0.105	0.659
SLAP sweep	–	–	1,794 (96.92)	5	2.793	9.895

primary evaluation: two previously recovered cases are unresolved under this study's limits. These outcomes are reported separately from verified exclusions. The paired timing cohort excludes 448 reactions because at least one configuration does not finish; the all-attempt cost column exposes the additional computational cost on those difficult reactions. Source pins, individual outcomes, directional costs, and reproduction scripts are provided with the code.

C.8 Archived end-to-end timing on coordinate reactions

The archived September implementation completed search and event-window decoding for 140/140 coordinate reactions. On this full set, the full pipeline takes a mean 6.07 CPU seconds per reaction (median 0.81; 95th percentile 25.47). Decoding accounts for a mean 4.65 CPU seconds (median 0.29). These measurements include search, final-family construction, decoding, and candidate output; they are separate from the search-only comparison above (Table A9).

The historical study measured one-seed, cap-2,000 cut-sweep search followed by its event-window decoder on the Apple M4 Pro CPU. CPU times are reported per reaction. The event thresholds and windows match the bond-event comparison, with no class-count cap. All 140 completed reactions reproduce the archived event-class sets and family counts. Median end-to-end wall time is 0.96 seconds. Timing includes input loading, checkpoint I/O, final-family construction, decoding, and candidate serialization; initial imports and reference-set verification are excluded. All 140 reactions finish within the five-minute watchdog. Detailed timing records and reproduction instructions are provided with the code.

Table A9 Archived MAPPA search plus event-window queries. CPU seconds per completed reaction in the 140-reaction pass. The final row measures the whole pipeline; medians and percentiles of individual stages are not additive. Completion coverage and timing scope are stated above.

Stage	Mean CPU s	Median CPU s	95th pct. CPU s
MAPPA sweep search	1.257	0.300	3.468
Final-family catalogue	0.156	0.019	0.595
Bond-event decoding	4.647	0.288	24.361
Candidate serialization	0.005	0.001	0.015
End to end	6.073	0.812	25.470